

AKSHAY PARALIKAR

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EDUCATION

New York University

09/2023 – 05/2025

Masters of Science, Electrical Engineering

GPA: 4.0/4.0

Relevant Coursework: Deep Learning, Advanced Python for Data Science, High-Performance ML, ML for Cyber Security, MLOPs

Veermata Jijabai Technological Institute

08/2016 – 09/2020

Bachelor of Technology, Electronics and Telecommunication

GPA: 8.11/10

SKILLS

Programming	Python, R, C, C++, SQL, PySpark, Hadoop, MATLAB, Numpy, Pandas, Scikit-Learn, PyTorch, TensorFlow, CUDA
ML Frameworks	SVM, k-means, XGBoost, LLMs, Time-Series Analysis, RNNs, LSTMs, Random Forest
Software Tools	Azure Databricks, Azure Data Factory, Excel, GitHub, PowerBI, Tableau, Grafana, Hive, Docker, Kubernetes

WORK EXPERIENCE

Fractal Analytics | *Data Science Consultant - Machine Learning* | *Mumbai, India*

09/2020 – 07/2023

- Built **Machine Learning Models** for price modeling for fast-moving packaged goods predicting annual sales using **Random Forest Regression** and NLP with **84%** accuracy
- Implemented **Bayesian Belief Networks** for ROI calculation of advertising investments across multiple platforms in R
- Conducted exploratory analysis and designed **ETL pipelines** on **Azure Data Factory** for data processing from ingestion to reporting, utilizing **Cloud Computing** and resulting in a **15% reduction** in overall processing time while enhancing reliability
- Employed advanced analytics on big data with **Python** and **PySpark** on **Azure Databricks** working on model building and created **CI/CD pipelines** for **model validation**, **model deployment**, and **cloud deployment** for **AI/ML** solutions
- Conducted **A/B testing** on model versions to evaluate performance differences and improve predictive accuracy
- Created interactive **dashboards**, **simulators**, and **optimizer** tools on **PowerBI** to visualize and forecast product recommendations and enable fact-based decision-making
- Built an investment **optimization tool** in **Tableau** for reshuffling investment amongst platforms and created a tool to calculate optimal time for investment to maximize ROI. Increased ROI by **7%**
- Achieved **Microsoft Certified: Azure Data Engineer Associate** certification, demonstrating expertise in Azure data solutions

Floodnet | *Sensor Data Engineer* | *NYC, USA*

04/2024 – 12/2024

- Created a comprehensive NYC-wide flood sensor health monitoring autonomous system utilizing time series data analysis for **200+ IoT flood sensors** for a \$7 million city-wide flood sensing initiative. Reduced downtime by **30%**
- Devised **filters** and **machine learning models** for noise filtering and accurate flood detection improving accuracy by **6%**
- Designed and optimized **SQL** queries in **pgAdmin** to prepare time series data tables for **data visualization** and analytics, enabling better decision-making for flood monitoring

PROJECTS

Efficient Vision Transformers: Once-Train Optimization: *New York University.*

09/2024 – 12/2024

- Developed a once-trained (OT) network framework for **Vision Transformers (ViTs)** to enable efficient deployment across diverse hardware platforms, utilizing **FlashAttention**, **Neural Architecture Search**, **Pruning**, **Distillation**, and **Quantization**
- Conducted in-depth **profiling** of **GPU-accelerated applications** to identify performance **bottlenecks**, memory usage patterns, and computational inefficiencies along with **benchmarking** methodologies to measure **inference performance**
- Achieved a model with **95%** fewer parameters and **75%** lower memory used with minimum performance tradeoffs on CIFAR-100

Jailbreaking Large Language Models: *New York University.*

09/2024 – 12/2024

- Developed **jailbreaking techniques** for large language models (LLMs) using the **Kov attack** method, leveraging **Monte Carlo Tree Search (MCTS)** to optimize adversarial prompts
- Evaluated the impact of key hyperparameters including depth, top-K sampling values, and moderation score aggregation methods on the success of jailbreaking **GPT-3.5**

Continual Learning for Autonomous Vehicles: *New York University.*

01/2024 – 05/2024

- Developed a **CNN continual learning** system for autonomous vehicle steering control using lifelong learning techniques resulting in the model adapting to new data without forgetting previous knowledge
- Formulated a novel method of **Temporal Consistency Regularization** with a balanced buffer to enhance prediction accuracy along with smoothness of drive reducing MSE by **18%** while also reducing memory used by **25%**

Optimized Graph Neural Networks for Credit Card Fraud Detection: *New York University.*

01/2024 – 05/2024

- Implemented and optimized a **Temporal Graph Attention (TGAT)** mechanism for semi-supervised credit card fraud detection, achieving an **F1-score of 0.88** on the Amazon fraud dataset
- Developed an architecture combining categorical and numerical attributes, using **multi-head attention** for neighbor importance and **attribute-driven gated residuals** to enhance interpretability and performance
- Reduced data preprocessing time by **90%** and model training time by **22%** using **cuda-numba** and **cython**